

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (Mechanical) (Credit System) Degree Examinations**

May - June 2023

20MEE24 – INTRODUCTION TO AIRCRAFT DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Any Missing data can be assumed suitably.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Differentiate between Airline and aircraft industries. Who are the major players in the industry?	10	L*2	1	1
b)	Briefly describe the functions of following parts in an aircraft i. Spars ii. Ribs iii. Stringers iv. Skin v. Empenage	10	L1	2	1
2. a)	What do you mean by drag co-efficient? Explain the different types of drag components	10	L2	1	1
b)	What do you mean by aircraft Mach number? Classify the flow based on Mach number.	10	L2	2	1
3. a)	Elucidate the characteristic features of shock waves.	10	L3	1	1
b)	Briefly explain and define the basic nomenclature of an aerofoil.	10	L1	2	1
Unit – II					
4. a)	Briefly explain the following: i. Indicated Air speed ii. Calibrated Air speed iii. True Speed iv. Equivalent Air speed	10	L1	3	1
b)	Write a short note on longitudinal stability and lateral stability	10	L1	4	1
5. a)	Write a brief note on Inverted maneuvers and maneuverability.	10	L1	3	1
b)	List the salient features of aerobatics in the action of loop and spin with examples.	10	L2	4	1
6. a)	Briefly explain correct and incorrect angles of bank.	10	L1	3	1
b)	With the help of graphs explain the effect of changes in engine thrust and altitude on power curves.	10	L1	4	1
Unit – III					
7. a)	List the different components in a landing gear system	10	L1	4	1
b)	What are the different types of landing gear configuration available today?	5	L1	5	1
c)	Briefly explain the landing gear geometries generally used in aircrafts.	5	L2	5	1
8. a)	List and explain the functions of Avionics?.	10	L1	5	1
b)	Briefly explain the different methods of navigation	10	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
